

Adequacy Score Between Declared and Real Urgency in a Digital Smart Green Supply Chain

R&D BST — Digital Smart Green Supply Chain: Planning Score



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Abstract

Modern supply chains operating under the Physical Internet paradigm face a fundamental signal-quality gap: the urgency declared by operators often diverges from the physically computable real urgency. This leads to over-triage (resource waste) and under-triage (hidden rupture risk). In Supply Chain 4.0 contexts marked by volatility and multi-tier dependencies, no existing digital twin architecture fully corrects this cognitive bias while propagating it consistently across a supplier network.

This report summarises a R&D internship that proposes a cognitive adequacy score $A(t) \in [0, 100]$ to measure and correct the gap between declared operator urgency and a multi-block, DAG-propagated digital twin real urgency $U_r(t)$. The approach combines pairwise elicitation (AHP, Fuzzy Best-Worst Method) for declared urgency, a noisy-OR aggregation of six operational KPI blocks for real urgency, PROMETHEE II outranking and systematic shock simulation for network-wide criticality, and a calibrated event engine for keeping the underlying KPI state current. The framework is implemented in Python and operated as a persistent, multi-tenant digital twin exercised through a weekly-cadence serious game, with its adequacy signal validated empirically against recorded operational outcomes.

Résumé

Dans une chaîne logistique, l'urgence qu'un opérateur déclare pour une tâche ne correspond pas toujours à l'urgence réelle de la situation. Un coordinateur sous pression a tendance à sur-déclarer (« crier au loup »), ce qui sature les moyens de transport prioritaires et fatigue les équipes ; à l'inverse, un risque réel peut rester sous-déclaré et passer inaperçu jusqu'à provoquer une rupture. Ce mémoire de fin d'études présente le travail réalisé durant un stage de R&D à la Direction Innovation Numérique d'ALTEN sur ce problème d'écart entre perception et réalité opérationnelle.

La contribution principale est un score d'adéquation cognitive $A(t)$, qui compare mathématiquement l'urgence déclarée par un opérateur (recueillie par questionnaire structuré, méthode AHP) à une urgence réelle $U_r(t)$ calculée à partir de six indicateurs opérationnels indépendants (délai, capacité, performance, risque, coût, empreinte carbone), puis propagée sur l'ensemble du réseau de fournisseurs. Le score pénalise plus sévèrement le risque caché que la fausse urgence, conformément aux résultats de la théorie de la décision sur l'aversion aux pertes. Ce cadre a été implémenté dans un jumeau numérique complet, *SupplyScore*, opéré comme un serious game à cadence hebdomadaire, incluant un moteur d'événements calibrés, une simulation de criticité réseau, et un service de calibration comparant les prédictions du modèle aux incidents réellement survenus. L'ensemble repose sur une base logicielle testée en profondeur (environ soixante modules Python, plus de 1400 tests automatisés), développée selon un processus qualité en trois jalons documentés.

1 R&D Operation Context

1.1 Internal Research Context

1.1.1 Smart Green Supply Chain Programme

The Direction Innovation Numérique (DIN) of ALTEN conducts applied R&D at the intersection of digital twin technology, logistics optimisation, and sustainable transport.

The Smart Green Supply Chain programme (Ben Zakour, 2024) articulates three complementary pillars:

- **Lean:** Minimisation of waste (cost, delay, over-stock) through pull-flow logic and real-time scheduling.
- **Green:** Minimisation of carbon footprint per transport leg, aligned with the Global Logistics Emissions Council (GLEC) Framework v3.2 (Smart Freight Centre, 2023) and ISO 14083:2023 (International Organization for Standardization, 2023).
- **Predictive:** Anticipation of disruptions through probabilistic modelling and machine learning, consistent with the digital supply chain twin paradigm (Ivanov; Dolgui, 2020).

This programme places ALTEN-DIN within the broader Industry 4.0 movement (Kagermann et al., 2013), which leverages cyber-physical systems, Internet of Things (IoT), and digital twins to transform reactive supply chains into adaptive, self-organising networks. The Physical Internet framework (Montreuil, 2011), proposing open, mutualized logistics networks analogous to internet routing protocols, provides the theoretical backdrop for the supply chain simulation environment described below.

1.1.2 Project: DISCO Digital Twin

DISCO (Supply Chain Illustrator) is a serious game and digital twin developed by the DIN to simulate a complete supply chain for a fictitious satellite product. It involves 22 component suppliers in domains such as structural assemblies, optical systems, and embedded electronics (Baxas, 2025). Players assume the role of supply chain buyers and make decisions about supplier selection, transport routes, and factory allocation. The DISCO technology stack is summarised in Table 2.

The scoring engine applies Min-Max normalisation over the values of the current game session:

$$\text{ScoreCO}_2 = 1 + 9 \times \frac{\text{realCO}_2 - \text{CO}_{2,\min}}{\text{CO}_{2,\max} - \text{CO}_{2,\min}}, \quad (1)$$

The operational variables of the CO₂ scoring model are described in Table 1.

Table 1: Component definitions for the CO₂ scoring equation.

Component	Type	Description
ScoreCO ₂	Dependent variable	Normalized greenhouse gas score on a [1, 10] scale
realCO ₂	Input variable	Measured carbon emissions on the transport leg
CO _{2,min}	System parameter	Minimum historical or target emissions threshold
CO _{2,max}	System parameter	Maximum historical or target emissions threshold

with analogous formulae for cost and risk. The scale ranges from [1, 10], where 1 → best performance and 10 → worst.

To support the showcase and continuous improvement of mature supply chain capabilities, the DISCO digital twin incorporates several specialized modules developed in prior research cycles: The routing module (Baulain et al., 2024), the risk scorer (Kabouche, 2025), the CO₂ quantification methodology (Cordova, 2024), and the software integration framework (Auclin, 2025).

Table 2: DISCO technology stack and key parameters.

Component	Technology / Value
Backend API	Spring Boot + Java 21
Database	PostgreSQL (Azure)
Frontend	Angular (localhost:4200)
REST API (test)	https://test-disco.altenlabs.net/back/
Authentication	Player key / Azure AD SSO OAuth2
Scores computed	CO ₂ , Cost, Risk per supplier/itinerary [1–10]

1.1.3 Capitalisation of Prior ALTEN-DIN Research

This R&D operation is situated within a body of prior DIN research on the DISCO programme. These works form the organisational and thematic

context of the present operation; the system delivered here (SupplyScore, Section 4) is a standalone Python application and does not integrate with these modules yet — their coupling is a stated perspective (Section 5), not a delivered capability.

Baxas (Baxas, 2025) defines the DISCO software architecture, including the Time To Service (TTS)/Time To Recovery (TTR) formalisation.

Baulain and Dubarry (Baulain et al., 2024) developed the Physical Internet routing module of DISCO, providing transit times via the HERE Maps API (Vincenty distances, mandatory 4h30 Heavy Goods Vehicle (HGV) pause compliance) and the Itinerary Performance Index (Itinerary Performance Index (IPG)) normalised in $[0, 1]$.

Kabouche (Kabouche, 2025) developed the supplier risk scorer combining delay history, geographic vulnerability, and financial fragility into a $[1, 10]$ scale.

Cordova (Cordova, 2024) developed the CO₂ quantification module per route using the GLEC/VECTO methodology with log-normalised output $\hat{x}_{\text{CO}_2}$.

Mondelice (Mondelice, 2025) developed the Transport Management System interface providing real-time transit-time extraction.

1.2 R&D Operation Brief

1.2.1 Title and Objectives

The R&D operation is entitled: *“Adequacy Score Between Declared and Real Urgency in a Smart Green Supply Chain Digital Twin”*. Its four operational objectives are:

- O1.** Define a multi-block, DAG-propagated real urgency $U_r(t)$ and a cognitive adequacy score $A(t) \in [0, 100]$ to measure the gap between the human planner’s declared urgency U_d and the digital twin’s operational state.
- O2.** Develop and implement a network propagation and criticality-driven prioritisation layer, combining PROMETHEE II outranking with systematic shock simulation, to identify structurally critical nodes.
- O3.** Validate the adequacy-driven prioritisation logic empirically, through worked scenarios and a calibration service comparing the hidden-risk signal against recorded operational outcomes.
- O4.** Operationalise the framework within a persistent, multi-tenant digital twin exercised as a weekly-cadence serious game, integrating a calibrated event engine, explainability, export, and administrative hardening.

1.2.2 Research Questions

Four core research questions drive the scientific approach:

RQ1 (Multi-block Real Urgency) How can a physically and operationally grounded real urgency $U_r(t)$ be computed dynamically in a digital twin from heterogeneous KPI blocks, and propagated coherently across a multi-tier supply chain graph?

RQ2 (Cognitive Adequacy) How should an adequacy score $A(t)$ be formulated to measure the alignment between operator declared urgency U_d and computed real urgency $U_r(t)$, asymmetrically penalizing under-triage vs. over-triage, and can this scoring demonstrate predictive validity against real outcomes?

RQ3 (Network Prioritisation) How can pairwise elicitation (AHP, Fuzzy-BWM) be combined with outranking (PROMETHEE II) and shock simulation to identify structurally critical nodes?

RQ4 (Operational Validation) How can this cognitive-physical framework be operationalised and empirically validated within a persistent, multi-tenant digital twin exercised as a weekly-cadence serious game?

These questions define the scientific scope of this Bilan Scientifique et Technique (BST): **RQ1** addresses multi-block urgency computation and propagation, **RQ2** defines the adequacy score formulation and its empirical validity, **RQ3** addresses network-wide criticality and prioritisation, and **RQ4** addresses operational validation within the digital twin.

1.2.3 Document Structure

This document is structured as follows. Section 2 formalises the scientific problem and outlines the primary research challenges. Section 3 reviews the state of the art across thematic subsections. Section 4 details the R&D iterations, methodology, and experimental results. Finally, Section 5 evaluates the research questions and discusses the project’s evolution.

2 Scientific Problem and Obstacles to Overcome

2.1 Relevant Context and Motivating Observations

Modern supply chain operators face simultaneous multi-dimensional assessments (cost, deadline, risk, carbon) where most monitoring and planning tools process one criterion at a time (Pinedo, 2012; Chakraborty et al., 2020); the underlying coordination problems are NP-hard in the general case (Garey et al., 1979), which is precisely why practitioners fall back on judgment-based prioritisation. This work addresses the layer that comes *before* any planning or scheduling decision, and which the classical literature has not fully investigated: the urgency signal itself is systematically biased at the source and must be measured and corrected before it can safely inform any downstream decision (M. Zhu et al., 2018b; Rusou et al., 2020). Feeding this corrected signal into an automated scheduler is a stated perspective of the programme (Section 5), not part of the delivered scope. In a Physical Internet network (Montreuil, 2011), where logistics capacity is mutualized across actors, a biased urgency signal propagates and amplifies systemic misallocations. Over-triage (declaring an urgency higher than the physical situation warrants) starves other network tenants, whereas under-triage (declaring an urgency lower than warranted) creates invisible stock rupture risk that cascades downstream (Li et al., 2020; Ivanov; Dolgui, 2020).

While digital twin architectures and real-time decision support systems have demonstrated strong capabilities in operational monitoring, disruption detection, resource reallocation, and order synchronisation (Ding et al., 2023; Leung et al., 2022; Hrušovský et al., 2021; Karam et al., 2022; X. Zhu et al., 2025a), they do so primarily by computing the physical operational state of the system. To the best of the present review, existing systems do not appear to explicitly model the gap between a declared client urgency and a computed operational urgency; the client-stated signal is rarely compared directly to live operational proxies (such as multi-block KPI risk, lateness probability, and shortage exposure) within the same model. The core scientific novelty of this R&D operation lies in the formal comparison between these declared and computed urgencies, propagated coherently across a multi-tier network, motivating the introduction of an urgency adequacy layer to address this missing integration in digital operations architectures.

2.2 Formal Problem Statement

Let $U_d \in [0, 1]$ denote the declared urgency and $U_r(t) \in [0, 1]$ the real urgency computed from operational data in the digital twin. The core variables and

domains are summarized in Table 3.

Table 3: Core variables and domains.

Symbol	Meaning	Domain
t	Current time (project hours, ISO week granularity)	$\mathbb{R}$
$U_r(t)$	Physically/operationally computable real urgency	$[0, 1]$
U_d	Operator-declared urgency	$[0, 1]$
ε	Tolerance threshold (cognitive noise band)	$[0, 1)$

The mismatch between both signals is defined as

$$\Delta U(t) = U_d - U_r(t) \in [-1, 1]. \quad (2)$$

The components of this mismatch equation are explained in Table 4.

Table 4: Component definitions for the urgency discrepancy equation.

Component	Type	Description
$\Delta U(t)$	Dependent variable	Raw urgency mismatch signal at time t
U_d	Input variable	Urgency level declared subjectively by the human operator
$U_r(t)$	Dynamic variable	Urgency level computed operationally by the digital twin

A tolerance band $\varepsilon \in [0, 1]$ is introduced so that small deviations are not treated as operationally meaningful. In this setting, over-declaration corresponds to $\Delta U(t) > \varepsilon$, under-declaration to $\Delta U(t) < -\varepsilon$, and approximate agreement to $|\Delta U(t)| \leq \varepsilon$. Potential consequences of these mismatch conditions are summarized in Table 5.

Table 5: Urgency pathologies and potential consequences.

Condition	Interpretation	Potential consequence
$\Delta U(t) > \varepsilon$	Over-declaration of urgency (false urgency)	Excess expediting, avoidable cost, possible delay for other flows
$\Delta U(t) < -\varepsilon$	Under-declaration of urgency (hidden risk)	Hidden lateness or stockout risk, delayed intervention, downstream disruption
$ \Delta U(t) \leq \varepsilon$	Acceptable alignment	No material urgency mismatch

An asymmetric treatment of mismatch may be introduced through weights such as $\lambda_{\text{under}} \geq \lambda_{\text{over}}$ when the application context treats under-prioritisation as more harmful than over-prioritisation. This is motivated by operations literature showing that decision errors can generate unequal downstream consequences (Kim et al., 2020; DiwasSingh et al., 2020), and by scheduling literature that models differentiated tardiness penalties rather than uniform delay costs (Baker et al., 1982; Chauhan et al., 2022). However, the degree of asymmetry is treated here as a design and calibration choice to be validated in the target use case, rather than as a universal property.

2.3 Formalised Research Questions

- RQ1. Multi-block real urgency ($U_r(t)$):** How can a physically and operationally grounded real urgency $U_r(t)$ be computed dynamically in a multi-tenant digital twin from heterogeneous KPI blocks (temporal, capacity, performance, risk, cost, environmental), and propagated coherently across a multi-tier supply chain graph?
- RQ2. Cognitive adequacy formulation ($A(t)$):** How should an adequacy score $A(t)$ be formulated to measure the alignment between operator declared urgency U_d and computed real urgency $U_r(t)$, while asymmetrically penalizing under-triage vs. over-triage, and can this scoring demonstrate predictive validity against real operational outcomes?
- RQ3. Network prioritisation and criticality:** How can pairwise elicitation methods (AHP, Fuzzy Best-Worst Method) be combined with outranking aggregation (PROMETHEE II) and systematic shock simulation to identify which nodes of a supply network are structurally critical and warrant prioritised attention?
- RQ4. Operational validation within a digital twin:** How can this cognitive-physical framework be operationalised and empirically validated within a persistent, multi-tenant digital twin exercised as a weekly-cadence serious game, and what predictive validity does it demonstrate against recorded operational incidents?

2.4 Research Hypotheses

To structure the empirical evaluation of the proposed framework, three research hypotheses are formulated:

- Hypothesis 1 (Cognitive Adequacy Debiasing):** Explicitly monitoring the adequacy score $A(t)$ between human declared urgency U_d and digital

twin computed real urgency $U_r(t)$ serves to debias operator prioritisation judgments, mitigating the "mere urgency effect" (M. Zhu et al., 2018a) and preventing the over-allocation of mutualized transport resources.

Hypothesis 2 (Network Criticality Robustness): Combining DAG-propagated real urgency with PROMETHEE II outranking and systematic shock simulation identifies the nodes whose failure would cause the greatest downstream impact, producing a criticality ranking that remains stable and reproducible under repeated what-if perturbations of the network.

Hypothesis 3 (Calibrated Event Integration): A taxonomy of calibrated supply chain events, revised through Bayesian (Beta-Bernoulli) and exponential-smoothing update operators, can integrate heterogeneous operational shocks into the KPI state without obscuring the visibility of the individual urgency blocks that compose it.

2.5 Technical and Scientific Challenges

To evaluate the validity of the proposed framework, its performance is investigated against six scientific challenges (locks) identified in contemporary literature:

Lock 1: Urgency Aggregation Model Justification

Real urgency can be modeled from a single deadline-proximity signal or from an aggregation of heterogeneous risk blocks (slack, capacity saturation, performance, failure exposure, cost drift, emissions overshoot). A single-block, purely time-driven curve is analytically convenient but discards information available in a live digital twin. Conversely, any multi-block aggregation rule must be theoretically justified against simpler linear or averaging alternatives (K. Chen et al., 1992; K. Chen et al., 1996; J. Chen et al., 2025; Mills et al., 2010).

A noisy-OR aggregation, in which a single saturated block is sufficient to drive the aggregate urgency toward its maximum, is grounded in the probabilistic-inference literature on conditional independence and combined causal evidence (Pearl, 1988) and in probabilistic risk-inference models for supply chain systems (Zheng; L. Zhang, 2020a). As a complementary, purely time-driven reference, a bounded hyperbolic discounting curve in the spirit of finite-time singularity models (Johansen et al., 2001; Johansen et al., 2000) is retained here strictly as an auxiliary tool for generating realistic synthetic urgency trajectories for testing and calibration,

not as the production aggregation rule; its suitability for that narrower purpose is evaluated separately from the main model.

Lock 2: Measurement Reliability of Declared Urgency

Eliciting the declared urgency U_d via pairwise multi-criteria methods (AHP, BWM, Fuzzy BWM) introduces cognitive noise and ranking instability. These methods are sensitive to human inconsistency and rank reversal, meaning that declared urgency cannot be assumed as a deterministic ground truth (Afrasiabi et al., 2022; Triantaphyllou; Yanase, 2024; Vieira et al., 2020).

Consequently, operator judgments must be modeled as noisy, stochastic inputs (Stam et al., 1997). The framework addresses this by enforcing consistency ratio ($ConsistencyRatio(CR)$) validation gates and interval boundaries to absorb cognitive fluctuations.

Lock 3: Value-Added of the Adequacy Signal

Computing $\Delta U(t) = U_d - U_r(t)$ is necessary but not sufficient. The decisive scientific test is whether this adequacy signal improves the identification of operationally at-risk nodes beyond the raw variables already available to an operator (local KPI thresholds, unweighted urgency, or simple node degree) (Cohen et al., 2021; Bojko et al., 2021).

If the score does not demonstrably outperform these simpler baselines against recorded outcomes, it serves merely as an auxiliary decision metric rather than a primary planning signal. This is addressed by citing behavioral operations research (Ahmetoglu, 2025) and human-in-the-loop mediation models (Turner et al., 2021), which indicate that visualizing the gap between operator perception and physical system state reduces operational cognitive biases, and is tested empirically through the calibration methodology detailed in Section 4.

Lock 4: Composite Score Interpretability

Aggregating multiple criteria into a single composite score, whether the local urgency U_r across six KPI blocks or the outranking flow across a network of nodes, risks obscuring trade-offs and weakening decision interpretability. Composite-index literature warns that nominal weights do not always reflect real influence due to component correlations (Greco et al., 2018; Eby, 2016; Borgonovo et al., 2021).

The proposed framework mitigates this by exposing each block’s contribution individually on the operator explainability page and by keeping the aggregation weights ω_m configurable and auditable rather than hard-coded, so that nominal and effective weights can be compared and, where needed, realigned through global sensitivity analysis.

Lock 5: Scientific Maturity of Network Risk Propagation vs. Front-End

Disruption propagation on supply chain graphs and digital-twin-driven replanning are mature, well-studied fields (Sawik, 2013; Sawik, 2014; Sawik, 2016; X. Zhu et al., 2025b; Z. Zhou et al., 2023; W. Zhang et al., 2026). The contribution of this work is not the propagation mechanics themselves, but making the urgency architecture defensible by:

- constructing a credible U_d input via consistency-gated elicitation,
- computing a physically-grounded, multi-block $U_r(t)$, and
- evaluating whether the adequacy gap between the two provides operational utility beyond standard KPI-threshold baselines.

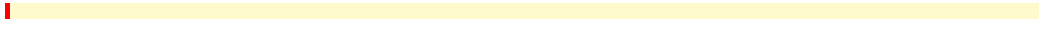
If only the propagation mechanism is delivered, the work represents a re-implementation of existing digital twin capabilities rather than a novel integration.

Lock 6: Absence of Digital Twin Precedent for Urgency Adequacy

Digital twin systems routinely monitor real-time operational states, disruption responses, and order synchronization (Ding et al., 2023; Leung et al., 2022; J. Zhang et al., 2021; Y. Zhang et al., 2026; C. Zhou et al., 2021). However, to the best of our knowledge, existing digital twin and decision support systems lack a module that simultaneously:

- captures a client-declared urgency signal,
- computes a model-based operational urgency from live system state, and
- measures their mismatch as an active, network-propagated indicator.

This three-layer architecture has no established precedent in the surveyed literature. Its integration feasibility and operational utility are investigated through this study to evaluate its practical viability and integration risks within a cognitive digital twin framework (Zheng; Lu, et al., 2021; Niloofar et al., 2023; Kalaboukas et al., 2021).



3 State of the Art

3.1 Introduction: Key Definitions

Urgency is defined here as the degree to which a supply chain task requires prompt action to avoid a material deterioration in operational performance. It differs from *importance*, which reflects strategic value, and from *priority*, which reflects the current execution order. In this work, urgency is treated as a multi-dimensional operational quantity, aggregated from heterogeneous KPI evidence rather than from a single deadline-driven signal (J. Chen et al., 2025; M. Zhu et al., 2018b).

Real-time scheduling refers to the class of algorithms that assign execution order to tasks given temporal constraints (deadlines, inter-task dependencies), resource constraints (vehicle capacity, supplier availability), and optimality criteria (makespan, weighted tardiness) (Pinedo, 2012). This literature is drawn on here only as historical grounding for deadline-sensitive value functions; the present work does not deliver a scheduling algorithm.

Digital twin is a synchronized digital representation of a physical system that supports monitoring, simulation, and decision-making using real-time and historical data (Ivanov; Dolgui, 2020). In supply chains, digital twins are used to improve visibility, disruption assessment, and replanning under changing conditions (Ivanov; Dolgui, 2020; Ding et al., 2023; Hrušovský et al., 2021).

Multi-criteria decision-making (Multi-Criteria Decision-Making (MCDM)) encompasses methods that combine multiple, potentially conflicting performance dimensions into a decision (Saaty, 1980; Brans; Vincke, et al., 1986). In the present work, the main dimensions considered for node prioritisation are time pressure, capacity saturation, operational performance, failure risk, cost drift, and carbon overshoot.

Physical Internet (Physical Internet (PI)) is the paradigm of open, mutualized logistics networks that route shipments through shared infrastructure using standardised modular containers (Montreuil, 2011). In such shared logistics environments, inaccurate urgency assessment may have effects beyond a single actor by influencing the allocation of mutualized capacity. This makes urgency evaluation particularly important in Physical Internet settings (Montreuil, 2011; Leung et al., 2022).

Serious game is a simulation environment designed to train decision-makers through realistic scenarios with immediate feedback and scoring (Sternman, 1989; Prensky, 2001). In this project, DISCO functions both as a serious game and as a digital twin: users interact with simulated supply chain decisions, while the underlying digital model evaluates operational consequences

on a weekly cadence.

3.2 Literature Review

The literature relevant to this framework is broad but fragmented. No retrieved work in the project searches combines, within a single supply-chain decision architecture, five elements at once: a multi-block operational urgency function, a structured client-side urgency input, an adequacy layer comparing stated and operational urgency, network propagation of that urgency across a multi-tier DAG, and empirical validation of the adequacy signal against recorded outcomes (Sawik, 2013; Sawik, 2014; Sawik, 2016; Cohen et al., 2021; Hosseini et al., 2020).

The review is organised by argumentative function. Two foundational streams establish *why* the problem exists. Two enabling streams provide the computational toolkit. One section covers the closest antecedents. Analogical literature provides rhetorical grounding but is labelled explicitly as non-foundational throughout.

Table 6 summarises the seven streams and their relationship to the thesis.

Table 6: Literature stream synthesis: contribution and gap relative to this thesis.

Stream	Role	Main contribution	Gap relative to this thesis
Risk propagation and deadline-sensitive value	Foundation	Monotone time-value functions; ripple-effect propagation on networks	No declared-vs-operational comparison
Cognitive bias in urgency	Foundation	Documents urgency mis-declaration patterns	No operational correction mechanism
Digital twins of network risk and MCDM supplier scoring	Closest antecedent	Multi-criteria supplier scoring and network disruption monitoring	No client-declared urgency or adequacy layer
Bayesian and probabilistic risk updating	Enabling method	Conjugate/probabilistic belief revision under new evidence	Not used to compare declared vs. computed urgency
Digital twin operations	Enabling method	Real-time replanning and disruption response	No stated-vs-operational urgency module found
MCDM methods	Enabling method	Criteria weighting and outranking	Rank reversal risk; nominal weights $\neq$ effective weights
Medical triage / PI / CO ₂	Analogy / Context	Asymmetric urgency vocabulary; network context	Analogical only (no formal transfer)

3.2.1 Stream 1: Risk Propagation and Deadline-Sensitive Value

Role: Foundation.

Why: It demonstrates that operational value can be formulated as a time-

dependent and network-dependent quantity, providing the primary motivation for treating urgency as a variable that is both time-varying and propagated across a multi-tier network rather than computed at a single node in isolation.

Within real-time systems, deadline-sensitive scheduling theory models task value as a function of completion or response time, a paradigm known as time-value or utility-accrual scheduling (K. Chen et al., 1996; J. Chen et al., 2025). Cheng and Mühlethaler (K. Chen et al., 1996) show that task utility is typically constant up to a deadline and then non-increasing with delay; the utility-accrual survey by Chen and Bella (J. Chen et al., 2025) confirms that deadline-sensitive literature represents task value as a function of response time with bounded, structured degradation after critical thresholds. This establishes the historical precedent for treating urgency as time-varying, but it stops at a single task in isolation.

A second, complementary line of evidence comes from network risk-propagation research. Ivanov and colleagues formalise the ripple effect, whereby a local disruption at one supplier cascades through downstream tiers and amplifies as it travels along the network (Ivanov; Dolgui, 2020; Ivanov; Dolgui, 2019; Li et al., 2020). Probabilistic risk-inference methods for supply chain systems (Zheng; L. Zhang, 2020a) further show that aggregating heterogeneous risk indicators into a single node-level score is itself an open modelling question, distinct from how that score should then be propagated.

Table 7 compares four candidate aggregation and growth models found across these two lines of literature.

Table 7: Comparative urgency aggregation and growth models and their properties.

Model	Formula	Key property	Use case
Linear (time)	$U_0 + \lambda t$	Constant pressure	Administrative tasks
Hyperbolic (time)	$K/(t_c - t)$, bounded via $\tanh$	Finite-time singularity, discounting	Synthetic scenario generation
Weighted average (blocks)	$\sum_m \omega_m u_m$	Fully compensatory: one weak block hidden by strong ones	Simple composite indices
Noisy-OR (blocks)	$1 - \prod_m (1 - u_m)^{\omega_m}$	Non-compensatory: one saturated block dominates	Multi-block operational risk

This thesis adopts the noisy-OR aggregation over the fully compensatory weighted average, on the grounds that a single catastrophic KPI block (for instance a complete capacity failure) should not be masked by acceptable scores on the remaining blocks, a property established in the probabilistic-inference literature on combining independent causal evidence (Pearl, 1988). The bounded hyperbolic discounting curve is retained separately, purely as an auxiliary tool for generating realistic synthetic urgency trajectories used in testing and calibration; a structurally similar acceleration appears in Log-Periodic Power Law (Log-Periodic Power Law (LPPL)) models of financial crash precursors (Johansen et al., 2001; Johansen et al., 2000), used here strictly as an analogical reference for that auxiliary generator, since empirical analyses by Feigenbaum (Feigenbaum, 2001) and Brée et al. (Brée et al., 2010) show that LPPL-family models are highly parameter-sensitive and unsuited to direct use as a production risk model.

Summary of Stream 1:

- **Established literature:** Deadline-sensitive scheduling supports time-dependent task valuation (J. Chen et al., 2025); ripple-effect research supports network propagation of local disruptions (Ivanov; Dolgui, 2020; Li et al., 2020).
- **Research gap:** Neither stream specifies how to aggregate heterogeneous, block-structured operational risk indicators at a single node before propagating them.
- **Working assumption:** A non-compensatory, noisy-OR aggregation of KPI blocks is proposed as the local urgency model, to be propagated with a leaky product rule across the network.
- **Systemic gap:** While these streams model network propagation and deadline sensitivity, neither addresses user-declared urgency or the mismatch between stated and operational urgency.

3.2.2 Stream 2: Cognitive Bias in Urgency Declaration

Role: Foundation.

Why: It establishes that client-declared urgency is systematically biased and cannot be taken at face value as a planning input.

Behavioural operations research suggests that human decision-makers systematically misrepresent urgency, a finding with direct implications for any system that relies on operator-declared urgency as an input.

Two convergent lines show this from different angles. Zhu, Yang and Hsee (M. Zhu et al., 2018b) report the Mere-Urgency Effect in five controlled experiments: participants consistently over-prioritised imminent low-value tasks over distant high-value tasks, and the bias persisted even when participants were informed of its existence. Steel (Steel et al., 2006) provides a complementary formal model through Temporal Motivational Theory (Temporal Motivational Theory (TMT)):

$$V(t) = \frac{U \cdot E}{1 + \Gamma \cdot D(t)^k}, \quad (3)$$

where U is objective utility, E is the expectancy of success, $D(t)$ is the delay to deadline, and Γ , k are individual discounting parameters. As $D(t) \rightarrow 0$, $V(t) \rightarrow U \cdot E$, so immediacy becomes more salient as the deadline approaches. Near-term tasks can therefore attract disproportionate attention even when their broader objective value is lower (Steel et al., 2006), which helps explain why imminent tasks may be over-declared as urgent. The operational roles of the parameters in this formulation are structured in Table 8.

Table 8: Temporal Motivational Theory (TMT) parameters and domains.

Symbol	Concept	Operational Role
$V(t)$	Temporal motivational value	Dynamic task salience
U	Objective utility	Task strategic importance
E	Expectancy of success	Probability of task completion
$D(t)$	Delay to deadline	Remaining safe execution window
Γ	Impulsiveness parameter	Operator sensitivity to delay
k	Sensitivity scaling parameter	Non-linear scaling of delay discounting

These laboratory findings are supported by evidence from operational settings. Rusou, Amar and Ayal (Rusou et al., 2020) show that operators under workload preferentially complete small, easy tasks, the operational analogue of under-triage for large-scope supply chain activities. DiwasSingh et al. (DiwasSingh et al., 2020) confirm in a large field study that completing easy tasks first significantly hurts overall performance. Hospital admission control under occupancy uncertainty (Kim et al., 2020) produces over- and under-triage patterns analogous to those in logistics operator decisions under uncertain lead times. The magnitude of these biases in industrial supply

chain contexts remains a subject of calibration. To mitigate these time-perception and workload biases, human-centered decision support systems (such as Rosenthal and Hiatt (Rosenthal et al., 2020)) use visual feedback techniques to show the mismatch between human plans and system temporal constraints, helping users calibrate their schedules.

This study treats declared urgency as a noisy signal and computes a model-based correction. The correction mechanism, and its empirical predictive validity, is detailed in Section 4.

Summary.

Known: Urgency over-declaration for near-deadline tasks is documented in controlled settings (M. Zhu et al., 2018b) and in operational settings (DiwasSingh et al., 2020; Rusou et al., 2020).

Unknown: Bias magnitude in industrial supply chain contexts.

Assumed: Asymmetric penalty $\lambda_{\text{under}} > \lambda_{\text{over}}$ is the appropriate correction structure.

Open gap: Behavioural research diagnoses the problem. It does not prescribe a real-time operational correction.

3.2.3 Closest Antecedents: Digital Twins of Network Risk and MCDM Supplier Scoring

Role: Closest antecedents.

Why: These are the works most structurally similar to the thesis. Identifying their gap precisely isolates what this work adds.

The works most structurally similar to this thesis combine multi-criteria supplier scoring with disruption-aware network monitoring. Sawik (Sawik, 2013; Sawik, 2014; Sawik, 2016) develops mixed-integer programming models for resilient supplier order assignment integrating cost, service level, and disruption risk. Cohen and Kaspi (Cohen et al., 2021) and Afrasiabi et al. (Afrasiabi et al., 2022) extend this to MCDM supplier prioritisation under disruption, combining criteria weighting with allocation optimisation. Chakraborty et al. (Chakraborty et al., 2020) confirm that rule-based heuristics remain competitive against exact methods in dynamic industrial environments. These works are cited here for the closeness of their subject matter (multi-criteria, disruption-aware supply chain decisions), not because this thesis reuses their optimisation machinery, which addresses allocation and sequencing rather than urgency measurement.

Taken together, this stream models disruption-aware supplier evaluation with multi-criteria weighting. Two gaps distinguish it from the present work. First, none introduces a network-propagated urgency signal computed

independently at every node and combined across tiers; risk is typically evaluated once, at the point of decision. Second, none compares a client-stated urgency against a model-derived urgency: the client-declared signal, where it exists, is used directly, without validation against live operational proxies.

Narrow gap statement. In the retrieved literature, no work simultaneously (i) aggregates heterogeneous operational risk into a multi-block urgency function propagated across a DAG, (ii) captures client-declared urgency as a structured, consistency-gated input, and (iii) feeds the mismatch between (i) and (ii) into an empirically calibrated adequacy score. Each stream addresses at most two of these three requirements.

Commercial positioning. The same gap was checked informally against commercial multi-tier supply chain risk-monitoring platforms (for example Everstream Analytics, Resilinc, and Interos), which map supplier networks to several tiers and continuously score them against pre-defined disruption scenarios. This is an informal product review rather than an academic source, and is stated here as the project’s own positioning assessment: to the team’s knowledge, none of these platforms compares a client-declared urgency signal against their own computed risk score, which is the specific mismatch this framework targets.

3.2.4 Enabling Method: Bayesian and Probabilistic Risk Updating

Role: Enabling method.

Why: Revising a failure probability from a small number of new observations is a component of operational risk state. Bayesian conjugate updating is the borrowed tool.

Bayesian methods have emerged as a well-supported approach for revising risk estimates from new evidence in supply chain research (Pearl, 1988; Koller et al., 2009). Rather than a full Bayesian network with an explicit conditional-probability-table structure, this study draws on the narrower and more tractable case of Beta-Bernoulli conjugate updating: a failure probability p is treated as a Beta-distributed belief with an equivalent prior sample size n_0 , revised from a small number of newly observed equivalent trials (k, n) each time a calibrated operational event is declared. This is the standard conjugate-prior construction for binomial-outcome belief revision (Pearl, 1988; Koller et al., 2009).

Three convergent findings from the retrieved Bayesian supply-chain literature support this choice of tool. Hosseini and Ivanov (Hosseini et al., 2020) provide the most comprehensive survey (259 citations) of Bayesian

methods applied to supply chain risk and resilience, finding that probabilistic belief-revision methods are often preferred to simple deterministic scoring when new, sparse evidence must update an existing risk estimate. Huang et al. (Huang et al., 2020) and Deng, Zhang and Wei (Deng et al., 2020) develop Bayesian models for disruption and delay probability estimation that serve as a methodological precedent for revising a KPI block’s failure probability from a declared event. Lockamy (Lockamy et al., 2011) and Qazi et al. (Qazi et al., 2017) further support Bayesian revision as a practical tool for dynamic supply chain risk quantification under sparse, event-driven evidence, consistent with the small-sample setting of a weekly-cadence digital twin. Plain frequency counts, by contrast, are unstable when only a handful of events are declared per node per project, which motivates the conjugate-prior formulation adopted here over a raw frequentist estimate (Zheng; L. Zhang, 2020b).

The specific conjugate-update parameters used for each event family are developed in Section 4.

Summary.

Known: Bayesian conjugate updating is an established, low-data-regime tool for revising failure probabilities in logistics contexts.

Unknown: The optimal prior sample size for DISCO’s specific event taxonomy.

Assumed: A shared, moderately informative prior (equivalent to roughly six months of weekly observations) is sufficient for initial calibration.

Open gap: Bayesian belief revision quantifies disruption exposure but does not, by itself, compare it against client-declared urgency.

3.2.5 Enabling Method: Digital Twin and Real-Time Decision Support

Role: Enabling method and primary gap source.

Why: DT systems are the architectural family closest to DISCO; this review establishes both what they already do and the specific gap this thesis addresses.

The digital twin concept was formalised by Grieves (Grieves, 2014) as a virtualised representation of a physical process. Tao et al. (Tao et al., 2018) extended the concept to manufacturing; Fuller et al. (Fuller et al., 2020) surveyed the enabling technologies underpinning modern deployments. In supply chain management, Ivanov et al. (Ivanov; Dolgui, et al., 2021) define the Supply Chain (SC) Digital Twin (DT) as integrating simulation (what-if analysis), optimisation (decision support), and machine learning (pattern detection). Ivanov and Dolgui (Ivanov; Dolgui, 2020) address the use of digital SC twins for disruption risk and resilience management in the Industry 4.0 era.

Wang, Wang and Liu (Wang et al., 2020) demonstrate DT-driven planning in a manufacturing context analogous to DISCO’s satellite component network. More recently, in the context of Industry 5.0, these systems have evolved towards Human-in-the-Loop (HITL) Cognitive Digital Twins (CDTs) where semantic models and reasoning methods are used to capture human feedback and synchronize mental models with simulated physical operations (Niloofar et al., 2023).

Real-time freight planning work demonstrates that operational urgency proxies (slack, lateness risk, and shortage exposure) are present within these systems (Hrušovský et al., 2021; Perez et al., 2022). Model-predictive control approaches for vehicle routing (Karam et al., 2022) and digital-twin disruption response (Ding et al., 2023; Leung et al., 2022; X. Zhu et al., 2025a) confirm the state of the art in live operational replanning.

Taken together, the digital-twin literature supports simulation, optimisation, and disruption-aware decision support. In the retrieved literature, however, no system surfaces a standard module that:

- captures a client-declared urgency signal as a structured input,
- computes a model-based operational urgency from the live system state, and
- measures their mismatch as an active, network-propagated indicator.

This three-layer architecture appears to be an open slot in digital SC operations (stated as an observation from the retrieved literature, not as a claim about the totality of published work). Whether filling this gap improves the identification of at-risk nodes is the central empirical hypothesis of this thesis (Lock 3, Section 2.5).

3.2.6 Enabling Methods: Multi-Criteria Decision Methods

Role: Enabling methods.

Why: These multi-criteria decision-making (MCDM) tools provide the computational foundation for criteria weighting, outranking, and network prioritisation.

This research integrates and builds upon two distinct methodological domains within multi-criteria decision-making: pairwise comparison methods for eliciting weights, and outranking aggregation for turning those weights into a defensible ranking of nodes.

Pairwise Elicitation and Pairwise Comparison Methods. To construct a structured representation of operator-declared urgency (U_d), criteria

weights must be systematically elicited. The Analytic Hierarchy Process (Analytic Hierarchy Process (AHP)) (Saaty, 1980) derives criteria weights as the principal eigenvector of a pairwise comparison matrix, validating logical consistency via the Consistency Ratio ($CR \leq 0.10$). While widely applied in logistics, AHP suffers from rank reversal, where the introduction of a new alternative alters the relative ranking of existing ones, and is sensitive to cognitive noise during expert elicitation (Dyer, 1990; Triantaphyllou; Yanase, 2024).

To minimize operator cognitive load and improve consistency, the Best-Worst Method (Best-Worst Method (BWM)) (Rezaei, 2015) reduces the number of required pairwise comparisons from $n(n-1)/2$ to $2(n-1)$ by anchoring comparisons around chosen "best" and "worst" criteria. This is further extended by the Fuzzy Best-Worst Method (Fuzzy Best-Worst Method (F-BWM)) (S. Guo et al., 2017; Yucesan et al., 2019; Amiri et al., 2020), which uses Triangular Fuzzy Numbers (Triangular Fuzzy Numbers (TFNs)) to absorb human judgment uncertainty. In this study, consistent FBWM weight vectors (Vieira et al., 2020) serve as an alternative elicitation gate to crisp AHP, and both feed the same downstream propagation and adequacy machinery.

Outranking Aggregation, Network Criticality, and Composite Indicator Critique. For prioritising which nodes require attention, direct weighted averages risk full compensatory aggregation, where an excellent score on one criterion (e.g., cost) completely masks a critical failure on another (e.g., capacity). To prevent this, outranking methods such as PROMETHEE II (Brans; Vincke, et al., 1986; Brans; Mareschal, 2005) establish pairwise preference flows (ϕ^+ and ϕ^-), allowing a non-compensatory ordering of alternatives. A comprehensive survey of PROMETHEE applications (Behzadian et al., 2010) confirms its suitability for supplier and node evaluation in industrial contexts. In this work, PROMETHEE II ranks active network nodes directly on the same weighted KPI blocks that feed the local urgency aggregation, giving a network-wide outranking view that complements the per-node adequacy score. Systematic what-if shock simulation, in which a node's local risk is forced to saturation and the resulting propagated delta is measured across the network, provides a second, independent criticality signal that is combined with the outranking flow to identify structural bottlenecks.

However, constructing any composite indicator from weighted criteria introduces methodological challenges. Composite-index literature warns that nominal weights assigned to criteria often diverge from their effective importance due to correlation, normalization, and aggregation dynamics (Greco

et al., 2018; Eby, 2016; Borgonovo et al., 2021). Furthermore, the choice of normalization method is non-neutral and can lead to rank reversal (Triantaphyllou, 2000). To keep the aggregation weights interpretable and defensible, this study exposes per-block contributions on the operator explainability page rather than only the aggregate score, while validating the risk parameters under uncertainty (Sawik, 2020).

3.2.7 Analogical and Contextual Literature

Role: Analogies and context.

Why: These bodies of literature provide conceptual analogies (medical triage), network contexts (Physical Internet), or regulatory frameworks (carbon accounting) that support the design of the adequacy variables.

While the core scoring and network-propagation methods are mathematically derived, their operational structure draws grounding and context from three non-foundational literature streams.

Medical Triage as a Conceptual Analogy. Medical triage provides a rich conceptual analogy for prioritizing tasks under severe capacity constraints and cognitive pressure. Standard protocols such as the Manchester Triage System (Manchester Triage System (MTS)) (Mackway-Jones et al., 2013) categorize patients into five urgency levels using flowcharts, whereas the Emergency Severity Index (ESI) integrates predicted resource consumption. Comparative clinical studies (Kuligowski et al., 2025) show significant classification discrepancies (weighted $\kappa = 0.51$) between these systems, highlighting that even trained professionals struggle with consistent urgency assessment.

In a supply chain context, this triage process maps directly to node prioritisation. The design of the cognitive adequacy score $A(t)$ is motivated by the clinical principle of asymmetric risk: under-triage (under-declaring urgency, leading to untreated critical conditions) is far more dangerous than over-triage (over-declaring urgency, leading to resource congestion) (Kim et al., 2020). Consequently, the adequacy score enforces an asymmetric penalty ($\lambda_{\text{under}} > \lambda_{\text{over}}$) to protect the physical supply chain from hidden stockout ruptures. Table 9 maps these triage concepts to their logistics equivalents.

Table 9: Heuristic mapping of medical triage concepts to DISCO logistics prioritisation.

Medical concept	DISCO logistics equivalent
Patient priority P1 (immediate)	Node with $A(t) < 20$ and high network criticality rank
Under-triage (P3 classified as P1)	Hidden risk $H = [U_r - U_d]_+$: silent bottleneck builds unnoticed
ESI resource consumption estimate	DISCO criticality index (propagated impact on final-customer nodes)
Resource-blind triage (MTS)	Local urgency threshold ignoring network position
Resource-aware triage (ESI)	Criticality-weighted prioritisation of coordinator attention

Physical Internet and Supply Chain Synchronomodality. The Physical Internet (PI) paradigm (Montreuil, 2011) conceptualizes open, mutualized logistics networks where routing, storage, and transport assets are shared. In such hyperconnected systems, urgency ceases to be a localized concern and becomes a network-level Quality of Service variable. Dynamic routing and order synchronization in synchronomodal networks require real-time, responsive priority signals compatible with rolling planning windows (W. Guo et al., 2020). Furthermore, disruptions propagate rapidly through hyperconnected networks, creating cascading ripple effects (Li et al., 2020) that validate the need for a synchronized, digital-twin-driven propagation model (Ivanov; Dolgui, 2020).

Regulatory Carbon Accounting. The integration of an environmental criterion is motivated by modern corporate sustainability requirements, such as the Corporate Sustainability Reporting Directive (CSRD Scope 3) and ALTEN’s environmental guidelines (Cordova, 2024). Standardized frameworks like the GLEC Framework v3.2 (Smart Freight Centre, 2023) and Vehicle Energy Consumption calculation TOol (VECTO) (European Commission, 2017) define emissions calculation baselines for heavy duty vehicles. ADEME (ADEME, 2014) reports that emissions factors are non-linearly related to vehicle speed and cargo load, supporting the treatment of carbon overshoot as its own urgency block rather than folding it into a generic cost term.

3.2.8 Validation Methodology: Serious Games

Role: Validation rationale.

Why: It justifies the weekly-cadence serious game used as the validation environment for the whole framework.

The Beer Game (Sterman, 1989) is the foundational supply chain serious game: Sterman demonstrates that players systematically under-react to upstream demand signals, generating the bullwhip effect. This decision bias is analogous to the under-triage pathology in logistics, where operators consistently under-respond to signals of building urgency. The finding that game-based observation captures systematic decision biases supports the use of digital twin serious games as an empirical instrument. Prensky (Prensky, 2001) establishes the theoretical basis for digital game-based learning, and Deterding et al. (Deterding et al., 2011) formalise gamification design principles.

Mohaddesi et al. (Mohaddesi et al., 2020) introduce the Gamette methodology, a playful, bite-sized approach to capturing decision-making patterns to build behavioural models, directly applicable to a weekly-review protocol in which operators repeatedly submit urgency assessments over the course of a project. Bellotti et al. (Bellotti et al., 2013) provide an evaluation framework for serious game learning outcomes, which supports the design of the calibration methodology detailed in Section 4.

3.3 Summary: Limitations of Existing Solutions

A review of the literature indicates that classical operational monitoring typically evaluates a single criterion (a deadline, a cost, or a risk score) at a time, failing to capture the multi-block nature of supply chain KPIs or the systemic impact of operator decision biases. While pairwise comparison methods (AHP, BWM, and Fuzzy BWM) are widely applied for criteria weighting, they function as static elicitation tools and are rarely compared against a computed operational counterpart. Outranking methods like PROMETHEE II prevent compensatory aggregation but are not usually combined with a network-propagated urgency signal.

Furthermore, while contemporary supply chain digital twins and Bayesian risk models offer strong capabilities in real-time disruption monitoring and belief revision, they typically lack a mechanism to compare human operator inputs with the physical state of the digital twin. This review identifies a key gap: the absence of an integrated architecture that compares a declared client-side urgency signal against a computed, network-propagated operational urgency to evaluate and correct cognitive mismatch, and that validates this correction empirically against recorded operational outcomes. Whether this

adequacy-driven prioritisation improves the identification of at-risk nodes compared to standard KPI baselines represents the central open question of this R&D operation.

3.4 Research Axis and Proposed Adequacy-Propagation Structure

To investigate this gap, this study proposes an architecture centered on a cognitive-physical adequacy check combined with network-wide risk propagation. The main objective is to design and evaluate a two-layer control loop:

1. **Cognitive adequacy check:** This layer captures the operator’s declared urgency U_d (elicited through consistency-gated pairwise comparisons) and compares it with a physically-grounded real urgency $U_r(t)$ computed from live digital twin state. The mismatch is processed using an asymmetric penalty to debias operator judgments.
2. **Network propagation and criticality-driven prioritisation:** Local urgency is propagated across the supply chain DAG (declared urgency descending toward suppliers, real urgency ascending toward the final customer), and the resulting network state feeds both a PROMETHEE II outranking ranking and a systematic shock-simulation criticality index, jointly identifying which nodes warrant prioritised operator attention.

This two-layer architecture is exercised and empirically validated within a persistent, multi-tenant digital twin operated as a weekly-cadence serious game, closing the loop between the adequacy signal and observed operational outcomes (Section 4).

4 Conducted Operations

4.1 Methodology: The Agile Test-and-Learn Program

This R&D operation is structured as an agile Test-and-Learn program designed to systematically investigate cognitive biases in supply chain urgency assessment and formulate a digital-twin-based correction loop. Rather than following a rigid sequential path, the program operates through continuous iterations of literature analysis, mathematical specification, software implementation, and scenario-based simulation.

The R&D pipeline is organized into five core operational areas:

- **Literature Review and Gap Analysis:** Establishing the psychological and mathematical foundations of multi-block urgency aggregation and cognitive bias (detailed in Section 3).
- **Mathematical Framework Specification:** Deriving and formalizing the multi-block real urgency function, its DAG propagation, and the asymmetric cognitive adequacy score from first principles to ensure dimensional and probabilistic consistency.
- **Pairwise comparison elicitation:** Formulating crisp Analytic Hierarchy Process (AHP) and Fuzzy Best-Worst Method (F-BWM) eigenvectors to build a reliable representation of declared operator urgency.
- **Software Implementation and Testing:** Constructing a modular Python library, exercised through a multi-page Dash web application, with comprehensive unit and property-based test coverage. The delivered codebase spans roughly sixty modules across eight packages (Table 16) and is exercised by over 1400 automated test functions, including property-based tests (Hypothesis library) that caught non-monotonic edge cases in the real urgency aggregation during development.
- **Simulation and Validation:** Conducting controlled shock and scenario simulations across a persistent digital twin, and validating the adequacy signal empirically against recorded operational outcomes through a dedicated calibration service.

4.2 System Architecture

Before the mathematical content, the delivered system’s architecture deserves a short description of its own: several design guarantees relied upon later (auditability, multi-tenant isolation, deterministic testing) are architectural, not mathematical.

Single facade. All operations go through one orchestrating facade (`services/orchestrator.py`) which coordinates the mathematical core, the graph repository, and the persistence layers behind a single thread-safe entry point. The web interface, the command-line entry point, and the test suite all drive the system exclusively through this facade — there is no second write path.

Multi-tenant persistence. Data is split between one global registry database (projects, nodes, arcs, milestones, scenarios, registry-level audit) and one SQLite database per node holding its private operational history (KPI snapshots, AHP assessments, weekly reviews, urgency history, declared events, and a per-node audit log). This isolation prevents write contention between operators reviewing different nodes and enforces the boundary that one supplier's raw history is not readable through another supplier's records.

Gatekeeper writes. Every business write crosses a single mutation service that whitelists the touched fields, validates ranges, diffs against the current state (a no-op write leaves no trace), records the change in the audit log with its author and source, and synchronises the in-memory graph. Audit tables are strictly immutable, including from the administration interface — the "no silent data loss" guarantee of Table 21 is enforced here, not by convention.

Graph and propagation. The network lives in an in-memory graph repository (networkx-backed; a Neo4j adapter is scaffolded but out of scope). Propagation is incremental: only the upstream/downstream cones of nodes whose local urgency actually changed are recomputed, which keeps the twin responsive at the 1000+-node scale validated in Milestone 3.

Decoupled clocks. Each project carries its own clock: a system clock for live use, a game clock advancing week by week for serious-game sessions, and a fixed clock for deterministic tests. The weekly cycle, milestone pressure, and event timestamps all read the project clock, never the wall clock directly — which is what makes a compressed serious-game campaign (Section 5) possible without touching the model.

4.3 Mathematical Framework and Core Formulations

Hypothesis. A multi-block noisy-OR urgency model, propagated across the supply chain DAG and compared against declared operator urgency through an asymmetric cognitive adequacy score, can resolve the discrepancy between human operator declared urgency (U_d) and digital twin computed

urgency ($U_r(t)$), providing a stable, explainable, and empirically calibratable prioritisation signal.

Approach. A systematic review of prior preliminary DIN formulations was conducted to identify mathematical inconsistencies (Lock 1, Lock 2, Section 2.5). The core physics of multi-block risk aggregation and cognitive bias correction were re-derived from first principles, establishing a two-layer control loop (cognitive adequacy check and DAG propagation with criticality-driven prioritisation) detailed below.

4.3.1 Core Model: Multi-Block Real Urgency and Network Propagation

Local Real Urgency ($U_{r,\text{local}}$). Real urgency is not read from a single deadline-proximity signal; it is aggregated at each node from six independent KPI blocks, {time, cap, perf, risk, cost, co2}, each producing a partial urgency $u_m \in [0, 1]$ or **None** when the underlying KPI data is unavailable. The aggregation is a weighted probabilistic OR, so that a single saturated block is sufficient to drive the local urgency toward its maximum, regardless of how well the other blocks score:

$$U_{r,\text{local}} = 1 - \prod_m (1 - u_m)^{\omega_m}, \quad (4)$$

where $\omega_m \geq 0$ is the aggregation weight of block m , and the product runs over blocks with available data. Table 10 summarises how each block is computed.

Table 10: The six real urgency blocks and their computation.

Block	Computation
u_{time}	Probability of lateness $P(L > d^* - t)$, with lead time L modelled as $\text{Normal}(\mu, \sigma)$; when an active milestone exists, additively modulated by its planning slippage $r = p_{\text{th}} - \text{progress}$, gained by κ_{retard} on delay and damped by κ_{avance} on advance.
u_{cap}	Weighted saturation of remaining volume, remaining weight, and unmet flow demand relative to capacity.
u_{perf}	$1 - \text{OEE}$ (Overall Equipment Effectiveness).
u_{risk}	$1 - \exp(-p_{\text{fail}} \cdot t_{\text{recov}} \cdot \text{sev}/T_{\text{ref}})$, composed by probabilistic OR with environmental and political exposure terms.
u_{cost}	Weighted operating cost drift, tariff multiplier excess, and storage cost relative to a reference cost.
u_{co2}	Normalised overshoot of measured emissions above a target, relative to a maximum threshold.

Two lifecycle status rules are applied before aggregation: a node marked **DONE** has its effective local U_r forced to 0.0 (a completed task carries no further physical risk), while a node marked **ABANDONED** has it forced to 1.0 (complete failure). These rules are applied identically whether the local urgency comes from the analytic blocks above or from a Monte Carlo estimate of the time block (Section 4.9).

Network Propagation. Declared urgency U_d propagates *downstream*, from the final customer (rank 0) toward raw-material suppliers, while real urgency U_r propagates *upstream*, from suppliers toward the final customer, using a leaky product rule that keeps both signals bounded in $[0, 1]$ and monotonically increasing as more neighbours become urgent:

$$U_{d,i} = 1 - (1 - U_{d,\text{local},i}) \prod_{k \in \text{Succ}(i)} (1 - \gamma_{ik} U_{d,k}), \quad (5)$$

$$U_{r,i} = 1 - (1 - U_{r,\text{local},i}) \prod_{j \in \text{Pred}(i)} (1 - \beta_{ji} U_{r,j}), \quad (6)$$

where $\gamma_{ik} \in [0, 1]$ is the downstream attenuation coefficient on arc (i, k) and $\beta_{ji} \in [0, 1]$ is the upstream amplification coefficient on arc (j, i) . Backup arcs are purely documentary and inert with respect to both propagations. Table 11 details the propagation variables.

Table 11: Component definitions for the network propagation equations.

Component	Type	Description
$U_{d,i}, U_{r,i}$	Dependent variables	Propagated declared/real urgency at node i
$U_{d,\text{local},i}, U_{r,\text{local},i}$	Input variables	Local urgency at node i after status-rule adjustment
γ_{ik}	System parameter	Downstream (demand) attenuation coefficient on arc $i \rightarrow k$
β_{ji}	System parameter	Upstream (risk) amplification coefficient on arc $j \rightarrow i$
$\text{Succ}(i), \text{Pred}(i)$	Structural sets	Nominal (non-backup) successors and predecessors of node i

Propagation is recomputed incrementally: only the nodes downstream or upstream of a change (a submitted questionnaire, a revised KPI, a structural edit) are recalculated, rather than the full graph, which keeps the digital twin responsive as the network grows.

4.3.2 Core Model: Asymmetric Cognitive Adequacy (Prospect Theory)

The mismatch between declared and real urgency is split into two named pathologies, false urgency F (over-declaration, panic) and hidden risk H (under-declaration, silent danger):

$$F = \max(U_d - U_r, 0), \quad H = \max(U_r - U_d, 0). \quad (7)$$

Drawing on Kahneman and Tversky’s Prospect Theory (Kahneman; Tversky, 1979), the two pathologies are penalised asymmetrically and with diminishing marginal sensitivity to large errors:

$$\text{penalty} = \lambda_{\text{under}} H^\alpha + \lambda_{\text{over}} F^\alpha, \quad (8)$$

and the penalty is rescaled into an adequacy score $A \in [0, 100]$ through an exponential mapping bounded by the penalty of a maximal one-sided error:

$$A = 100 \cdot \frac{\exp(-\text{penalty}) - \exp(-\text{penalty}_{\max})}{1 - \exp(-\text{penalty}_{\max})}, \quad \text{penalty}_{\max} = \max(\lambda_{\text{under}}, \lambda_{\text{over}}). \quad (9)$$

Table 12 lists the parameters, with their governing constraint $\lambda_{\text{under}} > \lambda_{\text{over}}$: a coordinator who panics wastes premium transport capacity, but a coordinator who under-reports risk misses the window to prevent a line stoppage, so under-declaration is priced more expensively.

Table 12: Component and parameter definitions for the asymmetric adequacy equations.

Component	Type	Description	Default
F	Dependent variable	False urgency (over-declaration)	—
H	Dependent variable	Hidden risk (under-declaration)	—
λ_{under}	System parameter	Penalty weight on hidden risk	2.25
λ_{over}	System parameter	Penalty weight on false urgency	1.0
α	System parameter	Psychophysical curvature (diminishing sensitivity)	0.88
A	Dependent variable	Adequacy score, 100 = perfect alignment	[0, 100]

The default $\lambda_{\text{under}}/\lambda_{\text{over}} = 2.25$ ratio reprises the loss-aversion coefficient measured by Kahneman and Tversky (Kahneman; Tversky, 1979; Kahneman,

2011), and $\alpha = 0.88$ reprises their measured curvature of the value function, so that a large mismatch is heavily penalised while the marginal penalty flattens for extreme errors.

Note on an auxiliary formulation. A separate, purely time-driven urgency function, $\tanh(K/\max(t_c - t, \varepsilon_{\min}))$, referencing the finite-time singularity literature (Johansen et al., 2001; Johansen et al., 2000), is also implemented in the codebase, but exclusively as part of the synthetic scenario generator used to produce realistic test and calibration data, not as the production computation of $U_{r,\text{local}}$ described above (Lock 1, Section 2.5).

Conclusion. The core model is mathematically consistent and dimensionally bounded on $[0, 1]$ (urgency) and $[0, 100]$ (adequacy) throughout. Both the aggregation weights ω_m and the penalty parameters ($\lambda_{\text{under}}, \lambda_{\text{over}}, \alpha$) are configurable per project rather than hard-coded, and their empirical calibration is addressed in Section 4.9.

4.4 Expert Criteria Weighting: The AHP Elicitation Model

Hypothesis. The Saaty AHP enables consistent weight derivation from expert pairwise judgements with $CR \leq 0.10$, producing a valid $U_d \in [0, 1]$ that encodes multi-dimensional operational urgency more reliably than a single intuitive score.

Approach. The Python module `core/ahp.py` implements the Saaty eigenvector method for a 4-criterion questionnaire administered through the weekly review page. The four criteria are:

- C1.** *Operational impact:* “If this task fails, what is the downstream consequence?”
- C2.** *Time window:* “How long before the situation becomes irredeemable?”
- C3.** *Downstream dependencies:* “How many tasks are blocked waiting for this one?”
- C4.** *Recoverability:* “Can we catch up if this task is late?” (inverted, where lower is worse)

The consistency ratio is computed as:

$$CR = \frac{\text{ConsistencyIndex}(CI)}{\text{RandomIndex}(\text{Saaty})n(RI)}, \quad (10)$$

where the components of the consistency assessment are defined in Table 13.

Table 13: Component definitions for the AHP consistency ratio equation.

Component	Type	Description
CR	Dependent variable	Consistency Ratio of the pairwise comparison matrix
CI	Dynamic variable	Consistency Index computed from the matrix principal eigenvalue
RIn	System parameter	Random Index for a matrix of size n (from Saaty’s reference table)

using Saaty’s Random Index table. If $CR > 0.10$, the interface flags the questionnaire as inconsistent and invites the operator to revise the most contradictory pairwise comparison before the assessment is accepted. Declared urgency is further smoothed with an exponential moving average ($\rho = 0.3$) across consecutive weekly assessments, absorbing transient stress spikes without erasing a sustained change in perception.

Results.

- Consistency validation, the eigenvector computation, and the EMA smoothing are exercised by the automated test suite, including property-based tests (Hypothesis library) that generate randomised comparison matrices to check that the consistency ratio and priority vector remain numerically stable across the input domain.
- Worked numerical traces (Section 4.6) confirm the eigenvector approximation, the consistency computation, and the resulting U_d against hand-verified reference values.
- Live, in-person expert elicitation campaigns with supply chain planners are identified as a validation step for a subsequent phase (Section 5) rather than a result already obtained.

Conclusion. AHP is viable for U_d encoding in the DISCO context. The CR validation gate, exercised by both worked examples and automated tests, successfully detects and rejects cognitively inconsistent expert inputs before they enter the network.

4.5 Stochastic Judgment Modeling: Fuzzy BWM and PROMETHEE II Network Outranking

Hypothesis. F-BWM TFNs weights provide an alternative, lower-effort elicitation route to crisp AHP under expert uncertainty, and PROMETHEE II

outranking, applied to the same weighted KPI blocks that drive local real urgency, produces a network-wide ranking of nodes that complements the per-node adequacy score.

Approach. The modules `mcda/fbwm.py` and `mcda/promethee.py` implement the two methods independently.

F-BWM Implementation. The linguistic TFN scale maps five qualitative judgements to triangular fuzzy numbers on a 1–4.5 scale (*égalent important* to *absolument plus important*), each carrying its own maximum consistency index. The consistency ratio of the Fuzzy BWM model is formulated as:

$$CR_{\text{BWM}} = \frac{\xi^*}{\text{CI}}, \quad (11)$$

where ξ^* is the optimal consistency indicator returned by a constrained non-linear optimisation (Scipy SLSQP) that fits the fuzzy weights to the elicited Best-to-Others and Others-to-Worst comparisons, and CI is read from the strongest linguistic judgement used. If the optimisation fails to converge, the implementation falls back to uniform weights rather than raising an error, so that a single malformed elicitation cannot crash the weekly review. Fuzzy weights are defuzzified with the Graded Mean Integration Representation (GMIR):

$$w_j = \frac{l_j + 4m_j + u_j}{6}, \quad (12)$$

then renormalised so that $\sum_j w_j = 1$. Table 14 defines the terms.

Table 14: Component definitions for the Fuzzy BWM defuzzification.

Component	Type	Description
w_j	Dependent variable	Crisp normalized weight of criterion j
l_j, m_j, u_j	Dynamic variables	Lower, modal, and upper bounds of the triangular fuzzy weight
ξ^*	Dynamic variable	Optimal consistency indicator from the fitted optimisation

PROMETHEE II Implementation. PROMETHEE II is applied in production to rank the active nodes of a project on the same six weighted urgency blocks (Table 10) that feed local real urgency, giving coordinators a network-wide, non-compensatory view of which nodes are collectively driving urgency. For each pair of active nodes (a, b) and each block m , the preference

intensity $P_m(g_m(a) - g_m(b)) \in [0, 1]$ is computed with a Type V linear preference function with an indifference threshold; the aggregated preference index and net outranking flow follow the standard PROMETHEE II formulation:

$$\pi(a, b) = \sum_m \omega_m P_m(g_m(a) - g_m(b)), \quad \phi(a) = \phi^+(a) - \phi^-(a), \quad (13)$$

where $\phi^+(a)$ and $\phi^-(a)$ are the average preference of a over, respectively by, all other active nodes. Table 15 defines the outranking flow terms.

Table 15: Component definitions for the PROMETHEE II net flow.

Component	Type	Description
$\pi(a, b)$	Dependent variable	Aggregated preference index of a over b
$\phi(a)$	Dependent variable	Net outranking flow representing overall preference ($\in [-1, 1]$)
$\phi^+(a), \phi^-(a)$	Dynamic variables	Positive (dominance) and negative (dominated) outranking flows

The same weights ω_m used in the local urgency aggregation (Equation 4) are reused as the PROMETHEE criteria weights, so that the network ranking and the per-node urgency score remain mutually consistent rather than relying on a separately calibrated weight set.

Conclusion. The F-BWM and PROMETHEE II modules are fully implemented, unit tested, and already wired into the production ranking path (`services/orchestrator.py`). F-BWM offers a lower-effort alternative to AHP for criteria weighting; PROMETHEE II turns those weights into a defensible, non-compensatory network ranking.

4.6 Worked Numerical Trace

Hypothesis. Declared urgency elicitation, multi-block real urgency aggregation, network propagation, and asymmetric adequacy scoring compose coherently end to end, producing traceable and explainable results at each stage.

Approach. A minimal three-node serial network is used to trace the full pipeline: Node 3 (raw material supplier) $\rightarrow$ Node 2 (sub-assembly) $\rightarrow$ Node 1 (final assembly, rank 0), with downstream coefficients $\gamma_{12} = 0.9$, $\gamma_{23} = 0.8$ and upstream coefficients $\beta_{21} = 0.7$, $\beta_{32} = 0.9$.

Step 1: AHP elicitation of U_d . An operator compares the four criteria of Table 10’s counterpart questionnaire, producing a reciprocal matrix whose priority vector is $w \approx (0.482, 0.157, 0.272, 0.088)$ with $\lambda_{\max} \approx 4.0145$, $CI \approx 0.0048$, and $CR \approx 0.0054 < 0.10$ (consistent). With local scores $s = (5, 3, 4, 2)$, the weighted score is $\sum w_j s_j \approx 4.148$, giving $U_{d,\text{current}} = (4.148 - 1)/8 \approx 0.394$. Smoothed against a previous value $U_{d,t-1} = 0.30$ with $\rho = 0.3$: $U_d = 0.3(0.30) + 0.7(0.394) \approx 0.366$.

Step 2: Local real urgency for each node. Node 1 is **ACTIVE** with $U_{r,\text{local},1} = 0.0$; Node 2 is **ACTIVE** with $U_{r,\text{local},2} = 0.40$ (aggregated from its own blocks); Node 3 is **DONE**, so its effective local urgency is forced to 0.0 by the status rule regardless of its raw block values.

Step 3: Network propagation. Descending U_d (topological order 1, 2, 3), with local declared urgencies $U_{d,\text{local}} = (0.80, 0.30, 0.50)$ (Node 3 keeps propagating U_d downstream despite being **DONE**, to preserve structural demand dependencies):

$$\begin{aligned} U_{d,1} &= 0.80 \\ U_{d,2} &= 1 - (1 - 0.30)(1 - 0.9 \times 0.80) = 1 - 0.7 \times 0.28 = 0.804 \\ U_{d,3} &= 1 - (1 - 0.50)(1 - 0.8 \times 0.804) = 1 - 0.5 \times 0.357 = 0.822 \end{aligned}$$

Ascending U_r (topological order 3, 2, 1):

$$\begin{aligned} U_{r,3} &= 0.0 \quad (\text{DONE}) \\ U_{r,2} &= 1 - (1 - 0.40)(1 - 0.9 \times 0.0) = 0.40 \\ U_{r,1} &= 1 - (1 - 0.0)(1 - 0.7 \times 0.40) = 0.28 \end{aligned}$$

Step 4: Asymmetric adequacy at Node 1. With $U_d = 0.80$ and $U_r = 0.28$: $F = 0.52$, $H = 0.0$. Using default parameters ($\lambda_{\text{under}} = 2.25$, $\lambda_{\text{over}} = 1.0$, $\alpha = 0.88$): $\text{penalty} = 1.0 \times 0.52^{0.88} \approx 0.567$, $\text{penalty}_{\max} = 2.25$, $\text{floor} = \exp(-2.25) \approx 0.105$, and:

$$A_1 = 100 \times \frac{\exp(-0.567) - 0.105}{1 - 0.105} \approx 100 \times \frac{0.567 - 0.105}{0.895} \approx 51.6.$$

Node 1 over-declares urgency relative to the network-propagated real risk; the score of $\approx 51.6/100$ reflects a moderate false-urgency penalty rather than a severe one, consistent with the asymmetric design where false urgency is priced less harshly than an equivalent hidden risk would be.

Conclusion. The full pipeline, elicitation, aggregation, propagation, and adequacy scoring, produces internally consistent, hand-verifiable results at every stage, which is a precondition for the explainability page described in Section 4.10.

4.6.1 Software Implementation Summary

The delivered codebase is organised in eight packages (Table 16), amounting to roughly sixty Python modules and over 1400 automated test functions, measured directly against the repository (`supplyscore/`) and cross-checked against the project’s code graph (5039 indexed code entities, 16517 call/reference edges across 183 files).

Table 16: Python packages of the SupplyScore implementation.

Package	Responsibility	Status
<code>core/</code>	U_r aggregation, AHP, adequacy, status rules, clock, per-node explainability	Implemented ✓, tested
<code>graph/</code>	DAG propagation, in-memory and Neo4j repositories, scenario/shock simulation	Implemented ✓, tested
<code>mcda/</code>	Fuzzy-BWM weighting, PROMETHEE II outranking	Implemented ✓, tested
<code>mc/</code>	Monte Carlo PERT lead-time simulation	Implemented ✓, tested
<code>services/</code>	Orchestration facade, weekly review, event engine, criticality, calibration, exports, admin	Implemented ✓, tested
<code>domain/</code>	Domain model, milestones, constraints, validation, event taxonomy	Implemented ✓, tested
<code>data/</code>	Multi-tenant SQLite persistence, migrations, backup, synthetic data generator	Implemented ✓, tested
<code>web_ui/</code>	Multi-page Dash application (dashboard, onboarding, weekly review, simulation, admin, explainability)	Implemented ✓, tested

Conclusion. The elicitation, aggregation, propagation, and adequacy layers behave as predicted on the worked trace above and are backed by a

substantially larger, independently tested codebase than a single illustrative example can convey.

4.7 Development Methodology and Quality Gates

Approach. Beyond the mathematical model itself, the delivered codebase followed a staged, gated development process rather than a single monolithic build, which matters for assessing how much confidence to place in the results above.

Three-milestone structure. The rewrite that produced the current system was organised into eighteen incremental phases, grouped into three milestones, each with an explicit exit condition (Table 17).

Table 17: Development milestones and their scope.

Milestone	Scope
Milestone 1 – “Ready for the serious game”	Onboarding wizard, ISO weekly cycle, calibrated event engine, bi-modal clock, universal versioning/audit, exports and backup.
Milestone 2 – “Analysis and defensibility”	End-to-end explainability, property-based proofs on the adequacy/event/milestone logic, safe free-form editing and admin view, prediction/reality calibration, F-BWM and PROMETHEE II weighting, Monte Carlo lead-time simulation.
Milestone 3 – “Industrialisation”	Performance at 1000+ nodes, multi-shock and criticality scenarios, systematic UI guardrails and automated browser testing, local packaging and hardened startup.

Quality gates. Each phase was only considered closed once it passed a fixed Definition of Done: static analysis clean (`ruff`, `mypy` strict on the core/graph/data/services packages), 100% of the test suite green, and a coverage threshold that started as a measured, non-blocking indicator in Milestone 1 and became a blocking gate at $\geq 90\%$ from Milestone 2 onward. The core governance parameters used throughout this report ($\lambda_{\text{under}} = 2.25$,

$\lambda_{\text{over}} = 1.0$, $\alpha = 0.88$, $CR < 0.10$, EMA $\rho = 0.3$, $\kappa_{\text{retard}} = 0.5$, $\kappa_{\text{avance}} = 0.2$, $n_0 = 26$) are treated as versioned constants: changing any of them is required to be a documented decision rather than a silent tuning change.

Early audit and iterative hardening. A code audit of the initial baseline (144 tests, no persistence hardening yet) catalogued thirteen concrete weaknesses before the rewrite began, including two rated critical: urgency state was not persisted across restarts, and SQLite connections created on the main thread were shared with Dash’s worker threads without the appropriate driver flag. These, along with several majors (duplicated status-rule logic in three places, double propagation on status changes, an unbounded per-node connection cache), were resolved across the following phases rather than left as known issues, which is the kind of iterative hardening a single end-to-end demo would not surface.

Performance and end-to-end validation. Milestone 3 added a dedicated performance phase validating the system at 1000+ nodes against measured budgets (`pytest-benchmark`, results archived with the test suite), and a UX/robustness phase adding eight automated Chrome (headless) browser journeys covering the main user flows, alongside a systematic guardrail that no raw exception is ever surfaced directly to the operator’s screen.

Conclusion. The adequacy and propagation results reported in this section were produced by a codebase developed under an explicit, auditable quality process (static analysis, blocking coverage, performance budgets, browser-level end-to-end tests), not only by unit tests of the mathematical core in isolation.

4.8 Weekly Review Cycle and Calibrated Event Engine

Hypothesis. A structured, signed, weekly review cycle, combined with a taxonomy of calibrated operational events, can keep a multi-tenant digital twin’s KPI state current without requiring free-form manual data entry, while preserving a complete audit trail.

Approach. The system is operated as a serious game advancing in discrete weeks (`core/clock.py` implements `SystemClock` for live use, `GameClock` for week-by-week advancement, and `FixedClock` for deterministic testing). At each new week, every active node is flagged as due for review, and the operator completes four sequential sections (`services/weekly.py`, served by the `/hebdo` page, Figure 1):

1. the AHP questionnaire (Section 4, pre-filled with the previous week's values, with a one-click "unchanged" confirmation);
2. the week's KPI values, shown against the previous week for quick differencing;
3. progress on active milestones, compared against theoretical planning progress;
4. any supply chain events that occurred during the week.

All four sections are written as a single signed, atomic transaction, so that no partial or anonymous review can corrupt the node's history.

Figure 1: Weekly review page: the four sequential sections (AHP, KPIs, milestones, events).

Calibrated Event Engine. Rather than asking operators to manually recompute the KPI impact of a disruption, the system exposes a taxonomy of fourteen calibrated event types (`domain/events.py`), listed in Table 18, each mapped to one or more KPI updates through one of three calibration operators.

Table 18: The fourteen calibrated event types.

panne_machine	retard_fournisseur	greve
hausse_tarif	rupture_matiere	accident
non_conformite_qualite	perturbation_transport	instabilite_politique
cyber_incident	hausse_energie	alerte_financiere_fournisseur
perte_capacite	pic_demande	

The three calibration operators are: a Beta-Bernoulli Bayesian update for failure probabilities,

$$p_{\text{post}} = \text{clip}\left(\frac{p \cdot n_0 + k}{n_0 + n}, 10^{-4}, 0.99\right), \quad (14)$$

with a prior weight $n_0 = 26$ (roughly six months of weekly observations); an exponential moving average for continuous variables such as OEE, $x_{\text{new}} = (1 - \lambda)x_{\text{old}} + \lambda x_{\text{obs}}$ with $\lambda \in [0.3, 0.5]$; and a ratchet (max) update for severity, $s_{\text{new}} = \max(s_{\text{old}}, s_{\text{event}})$. Table 19 defines the shared terms.

Table 19: Component definitions for the event calibration operators.

Component	Type	Description
p_{post}	Dependent variable	Revised failure probability after the event
n_0	System parameter	Prior equivalent sample size (26, roughly six months)
(k, n)	Event input	Equivalent failures/trials implied by the declared event's severity
x_{new}	Dependent variable	Revised continuous KPI (e.g., availability)
λ	System parameter	EMA reactivity weight, $\in [0.3, 0.5]$

Numerical Example: Strike Event. A node declares a **greve** (strike) lasting $d = 24$ hours affecting a fraction $p_f = 0.50$ of the workforce. Equivalent lost hours over a 168-hour week: $H_{\text{lost}} = d \cdot p_f = 12$, so $f_{\text{lost}} = 12/168 \approx 0.0714$ and the observed availability is $A_{\text{obs}} = 1 - 0.0714 \approx 0.9286$. Updating a baseline availability $A_{\text{old}} = 0.95$ with EMA reactivity $\lambda = 0.5$: $A_{\text{new}} = 0.5(0.95) + 0.5(0.9286) \approx 0.9393$. Risk severity is updated in ratchet mode: $S_{\text{new}} = \max(0.30, 0.50) = 0.50$. Operators can preview the transient impact of a declared event before confirming it, and reverting an event recomputes the baseline KPIs automatically.

Conclusion. The weekly cycle and event engine let coordinators keep the digital twin's KPI state current from structured, auditable inputs rather than free-form edits, which is a precondition for treating the resulting urgency history as a reliable calibration dataset (Section 4.9).

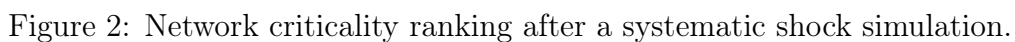
4.9 Monte Carlo Simulation, Network Criticality, and Empirical Calibration

Hypothesis. Replacing the analytic time block with a Monte Carlo estimate of delay probability, combined with systematic shock simulation and

PROMETHEE II outranking, identifies structurally critical nodes; and the resulting adequacy signal has measurable predictive validity against recorded operational outcomes.

Monte Carlo Lead-Time Simulation (PERT). `mc/lead_time.py` replaces the analytic u_{time} block with a full stochastic simulation when enabled, because the maximum of several lognormal lead times is not itself lognormal, which makes closed-form PERT approximations unreliable on multi-tier networks. Lead times are drawn from lognormal, normal (clipped to zero), or triangular distributions per node; completion times propagate topologically along the DAG, $S_i = \max_{j \in \text{Pred}(i)} C_j$, $C_i = S_i + L_i$; and the delay probability is estimated as $u_{\text{time}} = P(C_i > d_i) \approx \frac{1}{N} \sum_k \mathbb{I}(C_i^{(k)} > d_i)$ over $N = 10\,000$ draws by default. A memory guardrail caps the simulation to $N \times n \leq 5 \times 10^7$ across n nodes, and the resulting per-node delay probability feeds directly into Equation 4 through the `u_time_override` extension point, so the same aggregation and propagation machinery is reused unchanged.

Network Criticality Index. `services/criticite.py` answers a systematic version of the question a shock simulation answers manually: which node, if it failed completely, would cause the most damage to final-customer delivery? For every active node, the service forces its local real urgency to 1.0, re-runs the ascending propagation, and measures the resulting delta at the final-customer nodes ($\Delta U_{r,\text{final}}$) and across the whole network ($\Delta U_{r,\text{max}}$). Nodes are ranked by $\Delta U_{r,\text{final}}$ descending, then $\Delta U_{r,\text{max}}$ descending, then alphabetically; nodes already saturated at $U_r \geq 1.0$ yield a delta of zero and sink to the bottom of the ranking, since their risk is already realised in the baseline. Reusing the three-node network of Section 4.6, forcing Node 2 to $U_{r,2} = 1.0$ propagates to $U_{r,1} = 1 - (1 - 0.7 \times 1.0) = 0.70$, giving $\Delta U_{r,\text{final}} = 0.70 - 0.28 = 0.42$ and $\Delta U_{r,\text{max}} = 1.0 - 0.40 = 0.60$ (Figure 2).



Empirical Calibration of the Adequacy Signal. `services/calibration.py` directly addresses Lock 3 (Section 2.5): does the hidden-risk signal H have predictive value beyond a raw KPI threshold? For each recorded (node, week) pair, the historical hidden risk H is compared against negative outcomes observed in a subsequent window $[S+1, S+\text{horizon}]$ (default horizon: four weeks), a missed milestone, an abandoned node, or a critical event. The resulting confusion matrix yields precision, recall, and F1:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (15)$$

On a small illustrative sample of four observation points at threshold $H \geq 0.5$ (one true positive, one true negative, one false positive, one false negative), $\text{Precision} = \text{Recall} = F1 = 0.50$; the service explicitly flags any calibration run with fewer than 30 observation points as too small to draw statistically significant conclusions, and defers a well-powered confusion matrix to the accumulation of further weekly cycles (Figure 3).

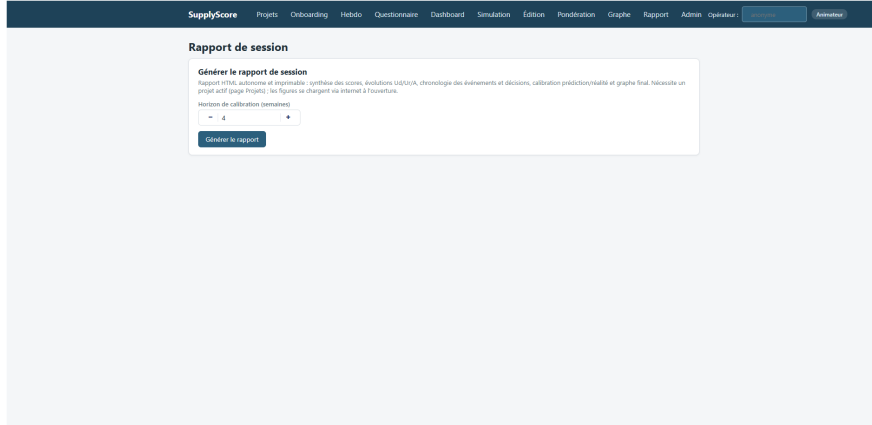


Figure 3: Calibration report: confusion matrix and precision/recall/F1 for the hidden-risk signal.

Conclusion. Monte Carlo delay estimation, systematic criticality ranking, and empirical calibration are implemented, tested, and wired into the same aggregation and propagation pipeline used elsewhere in the document. The calibration service in particular gives Lock 3 and RQ2 a concrete, falsifiable empirical test rather than a purely theoretical argument, subject to the sample-size caveat above.

4.10 Explainability, Export, and Operational Hardening

Approach. Three supporting capabilities complete the operational demonstrator.

Explainability (XAI). The per-node explanation page (`services/explain.py`) decomposes U_d into the exact linear contribution of each AHP criterion, $\kappa_j = w_j(s_j - 1)/8$ with $\sum_j \kappa_j = U_d$, and decomposes local U_r into the log-survival contribution of each KPI block, $\ell_m = -\omega_m \ln(1 - u_m)$, with $U_{r,\text{local}} = 1 - \exp(-\sum_m \ell_m)$ and per-block share $c_m = \ell_m / \sum_k \ell_k$, so that a coordinator can see precisely which block is driving a node's urgency and whether the mismatch reflects false urgency or hidden risk (Figure 4).

Figure 4: Criteria weighting interface (Fuzzy-BWM) feeding the explainability decomposition.

Export and Reporting. `services/exports.py` compiles global registry and per-node history into a twelve-worksheet Excel workbook (project meta-data, nodes, arcs, milestones, KPI history, urgency history, assessments, weekly reviews, KPI snapshots, events, decisions, and two audit-log sheets), converting all epoch timestamps to the coordinator's local timezone.

Operational Hardening. Because the persistence layer is SQLite, `data/backup.py` enables Write-Ahead Logging, runs `PRAGMA integrity_check` before every backup, and uses the native `sqlite3.backup` API to copy each database

without locking the running application; automatic backups run every 24 hours with a retention cap of 20 archives, and schema migrations are versioned through `PRAGMA user_version` (Figure 5).

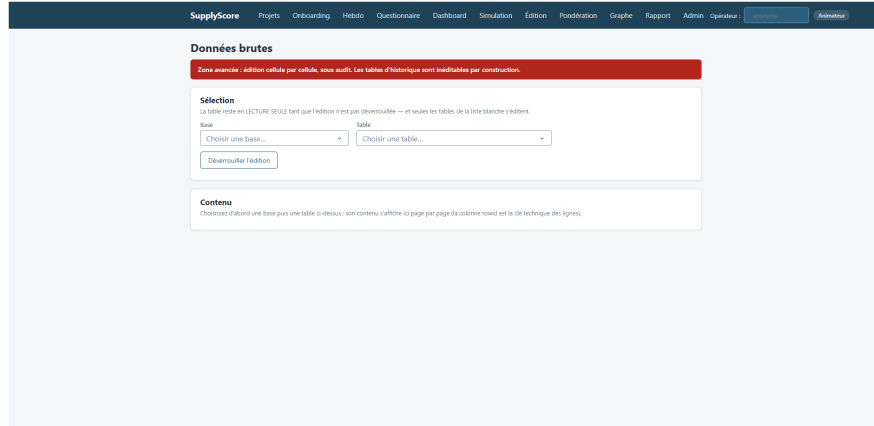


Figure 5: Administration page: database sizes, migration version, and backup controls.

Conclusion. Explainability, export, and hardening are not peripheral features here: they are what make the adequacy score auditable and the digital twin operable outside a development environment, which is a precondition for the empirical calibration described above.

5 Conclusions and Perspectives

5.1 Review of Research Questions

Status: Implemented and Tested (●)

The noisy-OR aggregation of six KPI blocks and the leaky DAG propagation of both U_d (descending) and U_r (ascending) are implemented, unit tested, and exercised by property-based tests. The worked trace in Section 4.6 confirms the model end to end on a hand-verifiable example. Remaining work concerns the empirical calibration of the block weights ω_m on a larger operational history than currently available (Lock 1, Lock 4).

RQ1: Multi-block real urgency $U_r(t)$ and network propagation.

Status: Implemented, Tested, and Empirically Calibrated on a Small Sample (●)

The asymmetric Prospect Theory adequacy score is implemented and tested, and its predictive validity against recorded operational outcomes is measured by a dedicated calibration service (Section 4.9) rather than asserted theoretically. On the sample collected so far, the service itself flags the observation count as below the 30-point threshold needed for a statistically significant conclusion; accumulating further weekly cycles to cross that threshold is the direct next step for RQ2 (Lock 3).

RQ2: Cognitive adequacy score $A(t)$ and its predictive validity.

Status: Implemented and Tested (●)

AHP and Fuzzy-BWM elicitation, PROMETHEE II outranking, and systematic shock-simulation criticality ranking are implemented, unit tested, and wired into the production ranking path. Live, in-person expert elicitation campaigns with supply chain planners, to validate the elicited weights against domain-expert judgment beyond synthetic and worked examples, remain an open next step (Lock 2).

RQ3: Network prioritisation and criticality.

Status: Implemented and Operational (●)

The framework is operated as a persistent, multi-tenant digital twin with a weekly review cycle, a fourteen-type calibrated event engine, explainability, export, and administrative hardening, all implemented and tested. The digital twin has been used to produce the worked and small-sample examples in Section 4; a longer-running, multi-session, multi-operator validation campaign is the natural extension to strengthen the statistical power of RQ2's calibration.

RQ4: Operational validation within a digital twin.

5.2 Research Challenges and Next Steps

Table 20 summarizes the status of the key research challenges (locks) and the proposed actions for their resolution.

Table 20: Research challenge status and target actions.

Challenge	Severity	Current Status	Target Action
Lock 1: Aggregation model	MEDIUM	Noisy-OR block aggregation implemented and tested, grounded in probabilistic-inference literature	Sensitivity study of ω_m against simpler aggregation forms
Lock 2: Elicitation noise	HIGH	Consistency ratio checks (CR) implemented in AHP and Fuzzy-BWM, tested	Run in-person expert elicitation sessions with supply chain planners
Lock 3: Adequacy utility	HIGH	Calibration service implemented; first confusion matrix computed on a below-threshold sample	Accumulate ≥ 30 observation points per project for a statistically significant validation
Lock 4: Weight interpretability	MEDIUM	Per-block explainability page implemented and tested	Formal variance-based (Sobol) sensitivity analysis of nominal vs. effective weights
Lock 5: Maturity contrast	LOW	Network propagation positioned against mature ripple-effect and digital-twin literature	Ongoing literature positioning as the contribution area is the adequacy layer, not the propagation mechanics

5.3 Limitations

The scope of the delivered framework was bounded by a set of explicit, documented design decisions rather than left implicit. Table [21](#) lists the limitations judged material enough to state directly, together with their operational implication.

Table 21: Documented limitations and their operational implications.

Limitation	Implication
No probabilistic guarantee	The framework produces probabilities, scores, and confidence intervals, not certainties; a model remains a simplification of the underlying reality.
Score freshness follows input cadence	Without an ERP connector, a node's score is never more current than its last submitted weekly review; manual entry is an accepted trade-off, not an oversight.
AHP is limited to four criteria	Saaty's method degrades past roughly seven pairwise-compared criteria, which is why declared urgency stays on four criteria while the larger set of KPI blocks is weighted through Fuzzy-BWM instead.
Event reversibility is conditional	Reverting a declared event is only guaranteed correct if no other write has touched the same KPI in the meantime; otherwise the system raises an explicit conflict for manual resolution rather than silently rebasing later writes, which would be mathematically dishonest.
Event calibration priors are defensible, not measured	The Beta-Bernoulli prior weight ($n_0 = 26$) and EMA reactivity coefficients follow standard conjugate Bayesian revision but are not empirically fitted; the serious game itself is the intended instrument for recalibrating them.
Local time urgency looks only at the next milestone	A distant milestone that is silently drifting stays invisible to u_{time} until it becomes the next active one; the underlying function already receives the full milestone list, so a weighted multi-milestone extension is possible without a model change.
Audit tables are strictly immutable	<code>audit_log</code> , <code>urgency_history</code> , <code>kpi_snapshots</code> , <code>weekly_reviews</code> , and <code>decisions</code> cannot be edited even from the admin view; relaxing this would undermine the "no silent data loss" guarantee the rest of the system relies on.
Double counting is possible	An operator can both declare a calibrated event and manually correct the same KPI in the same week; the interface warns but does not forbid this, since the human operator remains the final authority.

5.4 Future Research Directions

The following directions are identified as realistic extensions of the delivered framework, consistent with the architecture already in place:

- R.1. [HIGH – prepared, awaiting execution] Empirical validation campaign.** A full validation campaign has been designed, built, and rehearsed end to end (but *not* yet executed with human participants): a serious-game replay of the 2020–2022 semiconductor shortage over an 8-node network anchored on the real actors of the crisis, with real urgency driven by public time series (INSEE production/failures/price indices, WSTS worldwide billings, US producer prices) and declared urgency elicited from 8 consultants through the existing weekly AHP questionnaire. The protocol is pre-registered (six falsifiable hypotheses with numeric thresholds, blinding rules, placebo controls, sensitivity analyses), the data pack is frozen by hash, and five successive dry runs with synthetic operator profiles validated the pipeline and calibrated the scenario — surfacing, among other findings, that the network criticality index degenerates under saturation (realized risk is no longer shockable), a methodological result in its own right. Executing this campaign yields 144 real observations, crossing the 30-observation threshold required by the calibration service for a statistically meaningful confusion matrix.
- R.2. [HIGH] Multi-expert elicitation.** Run structured AHP and Fuzzy-BWM elicitation sessions with several supply chain planners per node, and aggregate their judgments with a geometric mean, consistent with the multi-expert aggregation path already anticipated in the elicitation modules.
- R.3. [MEDIUM] Global sensitivity analysis.** Conduct variance-based sensitivity analysis (Sobol indices) (Borgonovo et al., 2021; Sobol, 2001; Saltelli et al., 2010) on the aggregation weights ω_m and the adequacy parameters (λ_{under} , λ_{over} , α) to identify which have the greatest impact on the adequacy score and the criticality ranking.
- R.4. [MEDIUM] Coordinator behaviour pattern detection.** Apply temporal sequence models (Temporal Graph Networks, LSTMs) to the accumulated weekly urgency history to automatically classify recurring coordinator behaviours, such as a persistently panicking coordinator (frequent false urgency without operational delay) or a silent bottleneck (sustained hidden risk at a structurally critical node).
- R.5. [MEDIUM] Adequacy-driven scheduling.** The original programme ambition, deliberately kept *out* of the delivered scope: once the adequacy

signal is empirically validated, feed the debiased urgency into an active scheduling engine that reorders deliveries and supplier priorities. The present operation delivers the measurement layer this scheduler would consume; the scheduling algorithm itself (choice of priority rule, queue stability under stochastic lead times, integration with DISCO’s planning) is future work and should not be read anywhere in this report as an existing capability.

- R.6. [MEDIUM] Autonomous recommendation loop.** Formulate corrective actions (activating a backup arc, requesting expediting, adjusting a milestone) as a Markov Decision Process over the network state, and evaluate an agentic recommendation loop, using sandboxed shock simulation to estimate the expected adequacy improvement of each candidate action before proposing it to the operator.
- R.7. [LOW] Graph backend scale-out.** Evaluate migrating the in-memory graph repository to the already-scaffolded Neo4j repository implementation for projects whose node and arc counts exceed comfortable in-memory operation.

Explicitly descoped extensions. Distinct from the open research directions above, a further set of extensions was deliberately excluded from the current scope during planning, with the underlying model or interface already anticipating them so that resuming the work does not require a redesign:

- **Time-to-Recover / Time-to-Survive stress-testing** (Ivanov; Dolgui, 2020): the model already carries the recovery time, stock, and flow data such an analysis would need; deferred until after a full serious-game validation cycle.
- **Backup arc activation:** the data model (`ArcKind.BACKUP`) exists and is exposed in the UI, but arcs are currently documentary only; the switching formula (delay, capacity trade-off) is not yet implemented.
- **Neo4j graph backend end-to-end:** an adapter implementation exists but is frozen out of the test coverage requirement pending a concrete need for a scale beyond the in-memory repository.
- **ERP/file-based data import:** manual weekly entry is an accepted design choice for the current validation phase, not a placeholder for a missing connector.
- **Networked multi-operator access:** the system assumes multiple players on a single shared workstation, consistent with its serious-game deployment; concurrent network access was judged to reintroduce

distributed-systems complexity disproportionate to the current validation goals.

- **Weighted multi-milestone time urgency:** u_{time} already receives the full milestone list and could be extended to weigh all upcoming milestones rather than only the next active one (Table 21).

5.5 Proposed Methodological Framework Summary

In summary, this R&D operation delivers a structured methodological framework integrating cognitive debiasing with network-propagated supply chain risk assessment, operated as a persistent digital twin. The key elements of this delivered framework consist of:

1. **A non-compensatory, multi-block urgency model:** Aggregating six independent KPI blocks through a noisy-OR rule and propagating the result across a multi-tier supply chain DAG, rather than relying on a single deadline-proximity signal.
2. **An asymmetric cognitive adequacy scoring mechanism:** Translating the psychologically documented "mere urgency effect" and Prospect Theory loss aversion into a mathematical penalty function $A(t)$ that differentiates between over-declaration and under-declaration of task urgency, with its predictive validity tested against recorded outcomes rather than assumed.
3. **A tested, operational implementation:** A modular Python codebase of roughly sixty modules and over 1400 automated tests, integrating AHP, Fuzzy-BWM, PROMETHEE II, Monte Carlo lead-time simulation, DAG propagation, a calibrated event engine, and a multi-page Dash application, operated as a persistent, multi-tenant digital twin exercised through a weekly-cadence serious game.

5.6 Bilan Personnel et Compétences Mobilisées

5.6.1 Compétences techniques mobilisées

Cette opération de R&D a mobilisé des compétences dans quatre domaines distincts, dont l'articulation constitue l'essentiel de la difficulté technique du sujet :

- **Aide à la décision multi-critères :** élicitation par comparaisons par paires (AHP, Fuzzy-BWM), validation de cohérence, agrégation par surclassement (PROMETHEE II).

- **Modélisation probabiliste** : agrégation noisy-OR, simulation de Monte Carlo, révision bayésienne conjuguée (Beta-Bernoulli) en régime de données rares.
- **Algorithmique de graphes** : propagation sur graphe orienté acyclique multi-rangs, tri topologique, simulation de choc et calcul de criticité réseau.
- **Génie logiciel** : architecture d'un système multi-tenant en Python, persistance SQLite durcie (WAL, sauvegarde, migrations), interface web Dash, démarche qualité en jalons avec couverture de tests bloquante et tests de bout en bout automatisés.

5.6.2 Apprentissages et difficultés rencontrées

Section à compléter par l'auteur avant soutenance.

Cette sous-section doit rester personnelle et ne peut pas être rédigée à la place de l'auteur. Elle gagne à répondre concrètement à :

- Quelle a été la principale difficulté technique ou méthodologique rencontrée, et comment a-t-elle été surmontée ?
- Qu'est-ce qui aurait été fait différemment avec le recul ?
- Quelles compétences (techniques ou non) ont le plus progressé au cours du stage ?
- En quoi ce stage éclaire-t-il un projet professionnel ?

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Glossary

- $A(t)$ Adequacy score at time t , quantifying the asymmetric Prospect Theory penalty between U_d and $U_r(t)$: $A = 100 \times (\exp(-\text{penalty}) - \text{floor}) / (1 - \text{floor})$ with $\text{penalty} = \lambda_{\text{under}} H^\alpha + \lambda_{\text{over}} F^\alpha$, $\in [0, 100]$. 74
- F, H False urgency $F = [U_d - U_r]_+$ (unjustified panic) and hidden risk $H = [U_r - U_d]_+$ (silent danger), the two asymmetrically penalised pathologies underlying $A(t)$. 74
- U_d Declared urgency, a scalar $\in [0, 1]$ elicited from the human operator via an AHP or Fuzzy-BWM questionnaire, propagated downstream across the supply chain DAG through the leaky coefficients γ_{ik} . 74
- $U_r(t)$ Real (operationally computable) urgency at time t : a weighted probabilistic OR aggregation of six local KPI blocks (time, capacity, performance, risk, cost, CO2), $U_{r,\text{local}} = 1 - \prod_m (1 - u_m)^{\omega_m}$, propagated upstream across the supply chain DAG through the leaky coefficients β_{ji} . 74
- $\Delta U(t)$ Raw urgency mismatch: $U_d - U_r(t) \in (-1, 1)$; positive = over-triage (false urgency), negative = under-triage (hidden risk). 74
- $\Delta U_{r,\text{final}}$ Network criticality delta: the increase in real urgency propagated to final-customer nodes when a given node's local urgency is forced to saturation in a systematic shock simulation. 74
- ω_m Aggregation weight of KPI block m in the real urgency noisy-OR aggregation, reused unchanged as the PROMETHEE II criteria weight for network-wide node outranking. 74

Acronyms

- AHP** Analytic Hierarchy Process. 30, 74
- BST** Bilan Scientifique et Technique. 12, 74
- BWM** Best-Worst Method. 30, 74
- CI** Consistency Index. 40, 41, 74

CR Consistency Ratio. 17, 30, 40, 41, 42, 57, 74

CSRD Corporate Sustainability Reporting Directive. 74

DAG Directed Acyclic Graph. 74

DIN Direction Innovation Numérique. 9, 74

DISCO Supply Chain Illustrator (Digital Twin, DIN ALTEN). 74

DT Digital Twin. 28, 29, 74

F-BWM Fuzzy Best-Worst Method. 30, 74

GLEC Global Logistics Emissions Council. 9, 32, 74

HGV Heavy Goods Vehicle. 11, 74

IoT Internet of Things. 9, 74

IPG Itinerary Performance Index. 11, 74

KPI Key Performance Indicator. 74

LPPL Log-Periodic Power Law. 24, 74

MCDM Multi-Criteria Decision-Making. 20, 26, 29, 74

MTS Manchester Triage System. 31, 74

PI Physical Internet. 20, 32, 74

PROMETHEE Preference Ranking Organisation METHod for Enrichment Evaluation. 74

RI Random Index (Saaty). 40, 41, 74

SC Supply Chain. 28, 29, 74

TFN Triangular Fuzzy Number. 30, 41, 74

TMT Temporal Motivational Theory. 25, 74

TTR Time To Recovery. 11, 74

TTS Time To Service. 11, 74

VECTO Vehicle Energy Consumption calculation TOol. 32, 74

XAI Explainable Artificial Intelligence. 74